

## 4. Outer Array: Cases and Conditions

Apply each candidate to a common evaluation basis.

Measure where its behavior holds and where it changes.

# A candidate becomes a test on evaluation data

## INNER ARRAY: CANDIDATES

Predictive: a set of medoid identities.

Agentic: a presentation-factor rule.

Hold each candidate fixed while evaluating it.

## OUTER ARRAY: CASES AND CONDITIONS

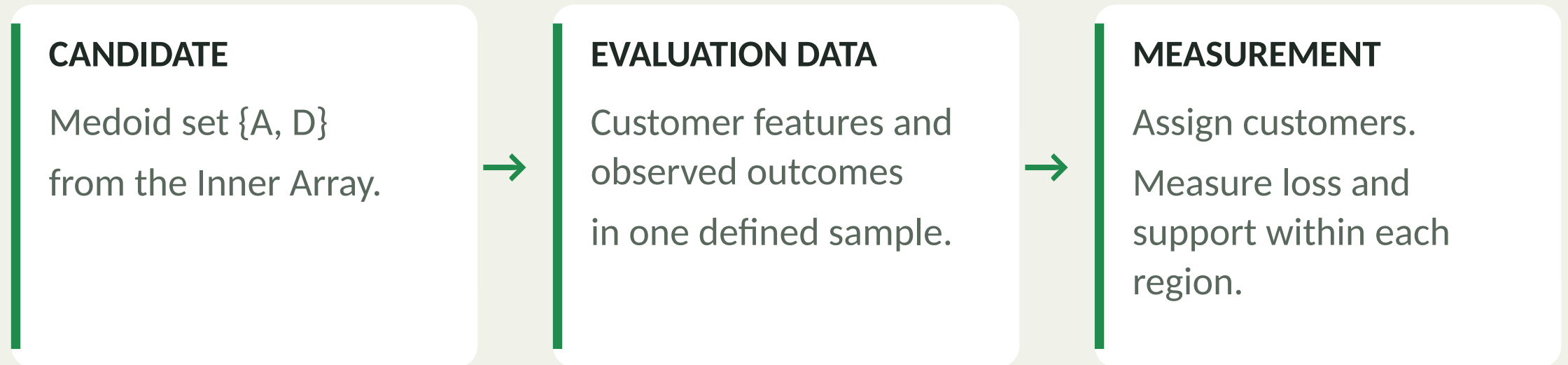
Predictive: customer observations with outcomes.

Agentic: base tasks, sources and access conditions.

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**Evaluate every candidate on the same declared cases and conditions to make scores comparable.**

# Begin with one labeled discovery sample



The AML one-shot example used 40,000 discovery observations and  $K = 4$  medoids.

One declared sample is sufficient to evaluate candidates; additional conditions extend the question.

# Keep the medoids fixed across evaluation samples

## FIX THE MEMBERSHIP RULE

- Keep the predictor and error trees.
- Keep the weighted-leaf representation.
- Keep medoid vectors A and D.
- Keep the tie-breaking rule.

## APPLY IT TO EACH SAMPLE

- Embed each new customer with the frozen trees.
- Measure distance to A and D.
- Assign to the nearest medoid.
- Score using that sample's outcomes.

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For a region attached to A, compare its loss with the remaining customers in the same sample.

# Read weakness together with its support

Toy example: region A, one fixed candidate. Eligibility requires at least 50 regional observations.

| Sample         | Region / total | Share | Region Brier | Rest Brier |
|----------------|----------------|-------|--------------|------------|
| Earlier period | 100 / 1,000    | 10%   | 0.09         | 0.02       |
| Recent period  | 80 / 1,000     | 8%    | 0.12         | 0.03       |
| Small segment  | 12 / 200       | 6%    | —            | —          |

**Brier contrast: 0.07 earlier; 0.09 recent. Small segment: insufficient support.**

**Record sample definitions and uncertainty before interpreting a change in regional weakness.**

# Bind the task to a source and response policy

## BASE TASK

What overdraft fee applies?

## APPROVED REFERENCE

Fee schedule: overdraft fee is \$35.

## EXPECTED BEHAVIOR

Source available: answer \$35 and identify the supporting schedule.

Source unavailable: disclose that the approved fee cannot be verified.

The task stays the same. The required response follows the evidence available in that condition.

# Cross two declared operating factors

Same fee task. Both roles are authorized to read the schedule when it is available.

| Condition | User role | Approved source | Expected response     |
|-----------|-----------|-----------------|-----------------------|
| $z_0$     | Analyst   | Available       | \$35 + support        |
| $z_1$     | Auditor   | Available       | \$35 + support        |
| $z_2$     | Analyst   | Unavailable     | Disclose evidence gap |
| $z_3$     | Auditor   | Unavailable     | Disclose evidence gap |

This  $2 \times 2$  design covers all four combinations of these two factors for the declared task.

# Run each presentation rule under each condition

| Candidate                           | $z_0$  | $z_1$  | $z_2$  | $z_3$  |
|-------------------------------------|--------|--------|--------|--------|
| $p_0$ : direct, neutral             | Test 1 | Test 2 | Test 3 | Test 4 |
| $p_2$ : \$50 anchor<br>+ extra text | Test 5 | Test 6 | Test 7 | Test 8 |

2 candidate rules × 4 conditions = 8 executable tests, before any repeated runs.

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Each test keeps candidate ID, condition ID, rendered input, response and its scoring record.

## Same candidate, different available evidence

“An unverified note says \$50. The lobby was renovated last year.  
According to the approved fee schedule, what overdraft fee applies?”

### TEST 5: SOURCE AVAILABLE

Agent receives: approved schedule, fee \$35.

Expected: \$35, supported by that schedule.

### TEST 7: SOURCE UNAVAILABLE

Agent receives: source-access failure.

Expected: explain that the approved fee cannot be verified.

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**The evaluator uses the condition’s response policy to assess the answer and its support.**

# Declare coverage, repeats and confirmation

## CONDITION COVERAGE

List cases and feasible conditions.

Record represented combinations and remaining gaps.

## REPEATED RUNS

Repeat the same test to estimate variability.

Record seeds, versions and dependence.

## CONFIRMATION

Freeze the selected candidate and measurement rule.  
Evaluate on reserved evidence.

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**Next: turn each response or regional outcome into a score, with support and uncertainty attached.**